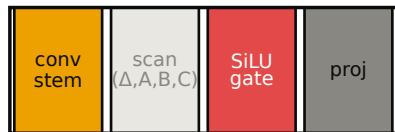


Step 1 -- Freeze the scan (necessity test)

conv stem scan (Δ, A, B, C) SiLU gate proj

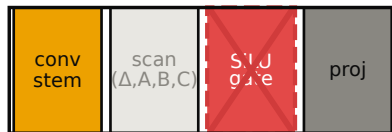
BM3-frozen



partial / full freeze vs S4D:
+12-18pp / +20-23pp (forward)
→ input-indep. coefficients suffice

Step 2 -- Remove gate (localise)

frozen – gate



partial host, forward: +9/+16/+27pp
(resolved at –6 dB only)
reverse: resolved at every level

Step 3 -- Add gate to S4D (sufficiency test)

S4D + gate
+ added



additive $\approx$ gateless (not assessable);
mult. – add. at –6 dB: +23 / +27pp
graft onto S4D: earlier protocol

Conditional on this block, testbed and budget: input-independent scan coefficients suffice; gate cost grows with noise only in the forward fully frozen block.